

Decision trees

Terminologies related to Decision Trees

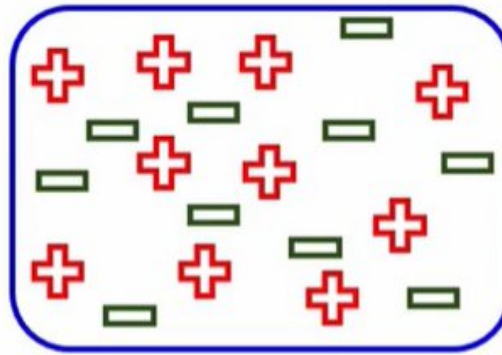
Different splitting criterion in Decision Trees

Optimizing performance of DT

DT used for classification and regression. Ensemble algorithm like random forest and XGBoost uses multiple DT

How to select best split can be decided by using techniques like gini impurity , CHAID , reduce in variance , CART

- Height
- Performance in Class
- Class

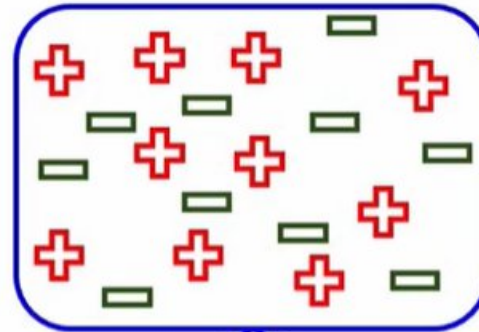


Students = 20

Play Cricket = 10

Percentage = 50%

- Split on Height



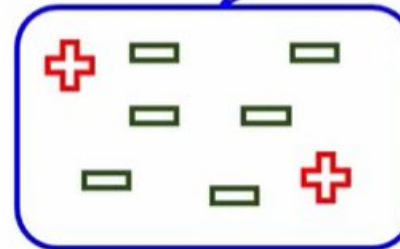
Students = 20

Play Cricket = 10

Percentage = 50%

<5.5 ft

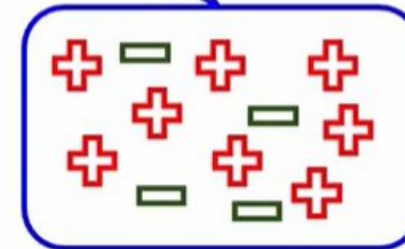
>=5.5 ft



Students = 8

Play Cricket = 2

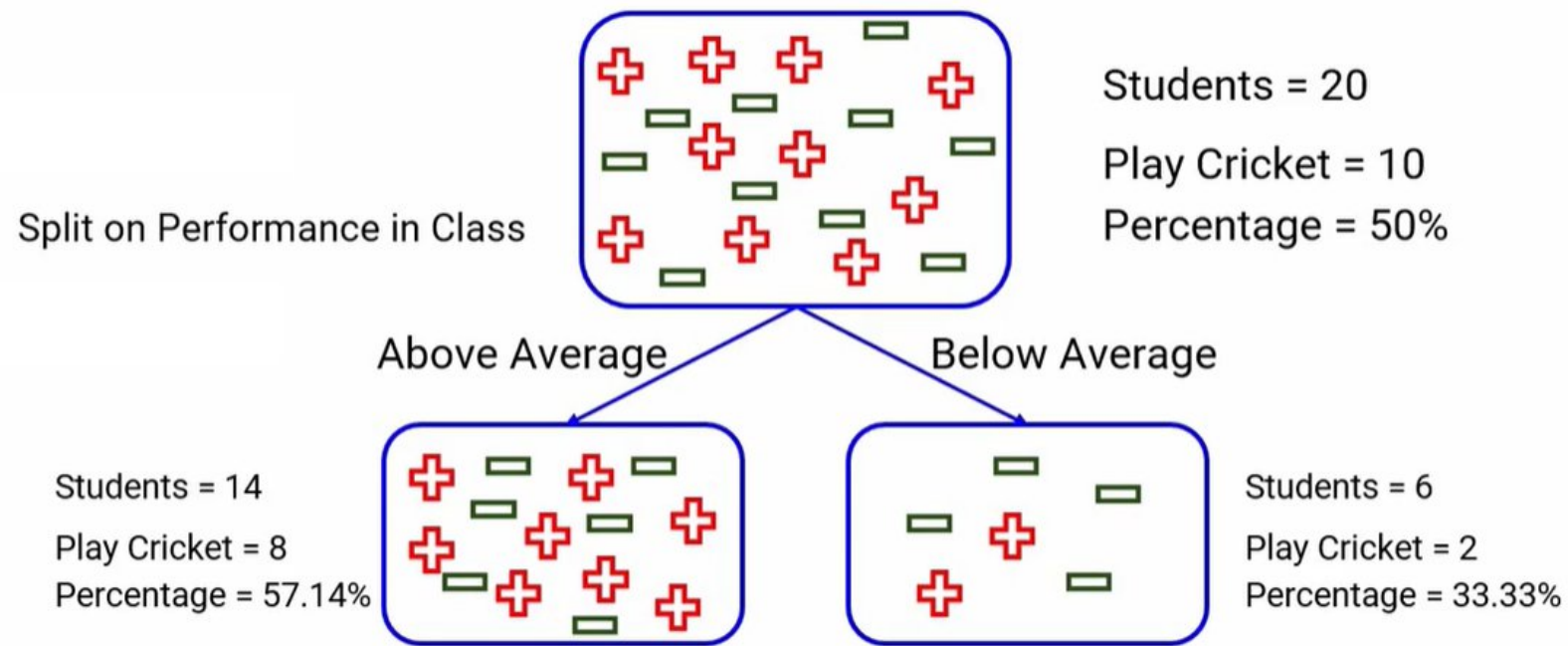
Percentage = 25%



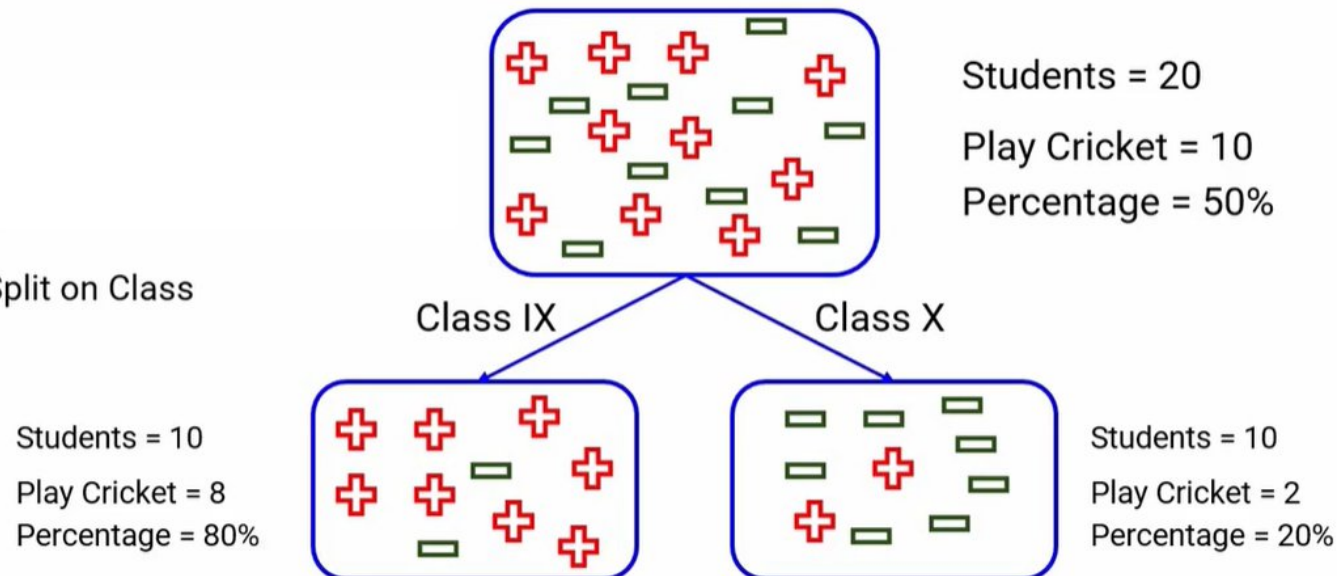
Students = 12

Play Cricket = 8

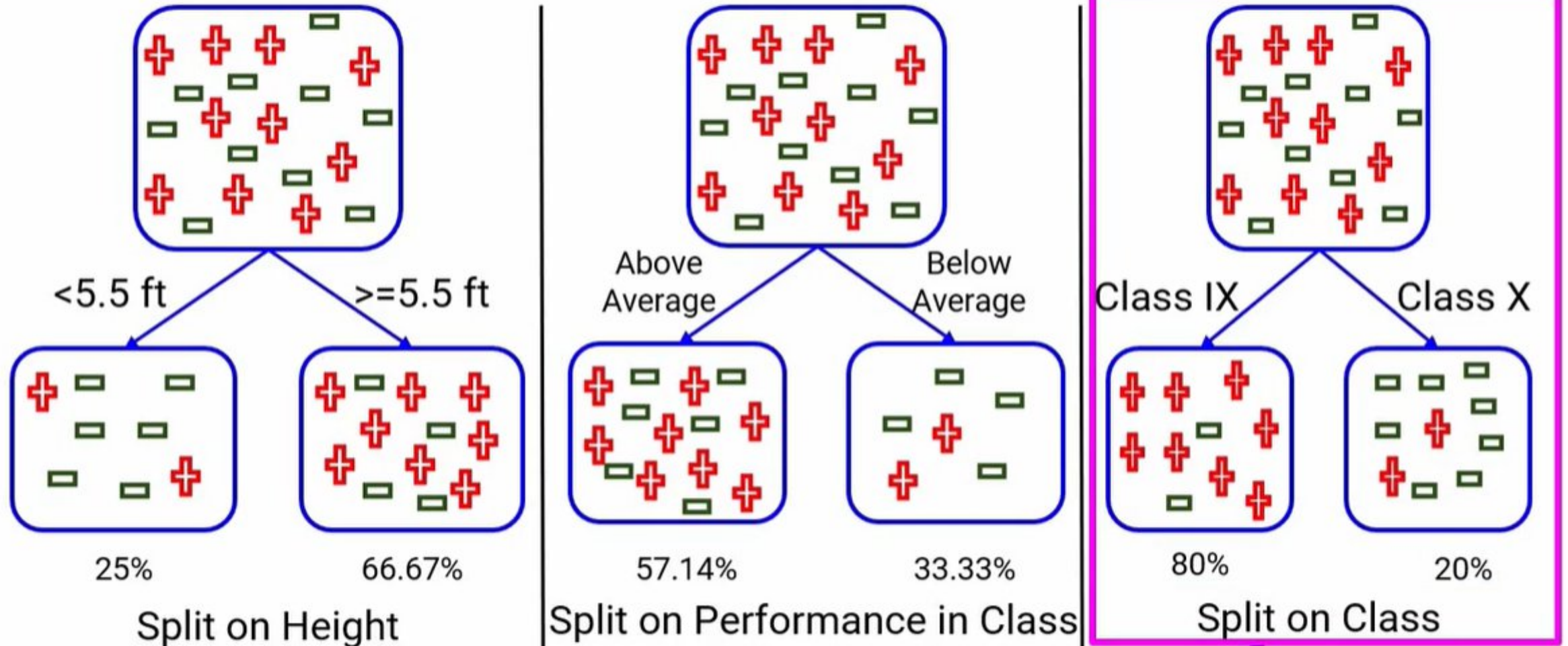
Percentage = 66.67%



- Split on Class

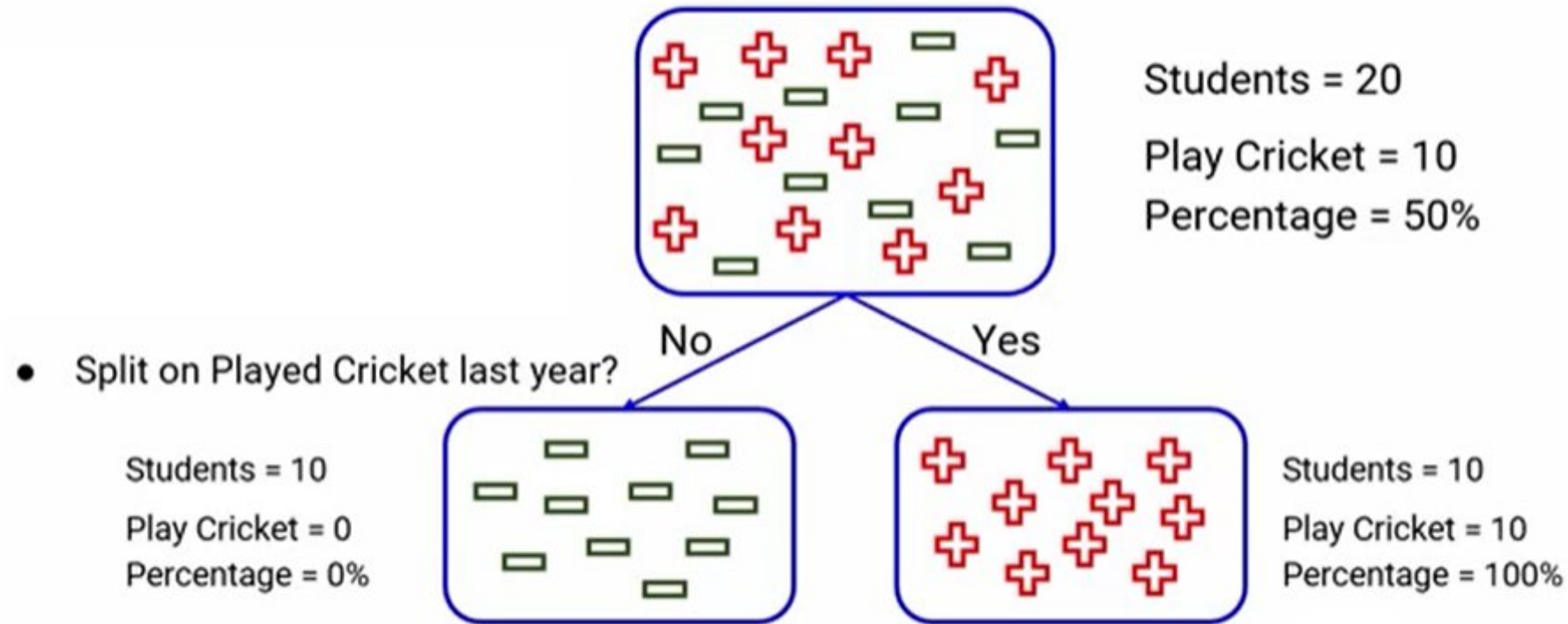


Pure nodes



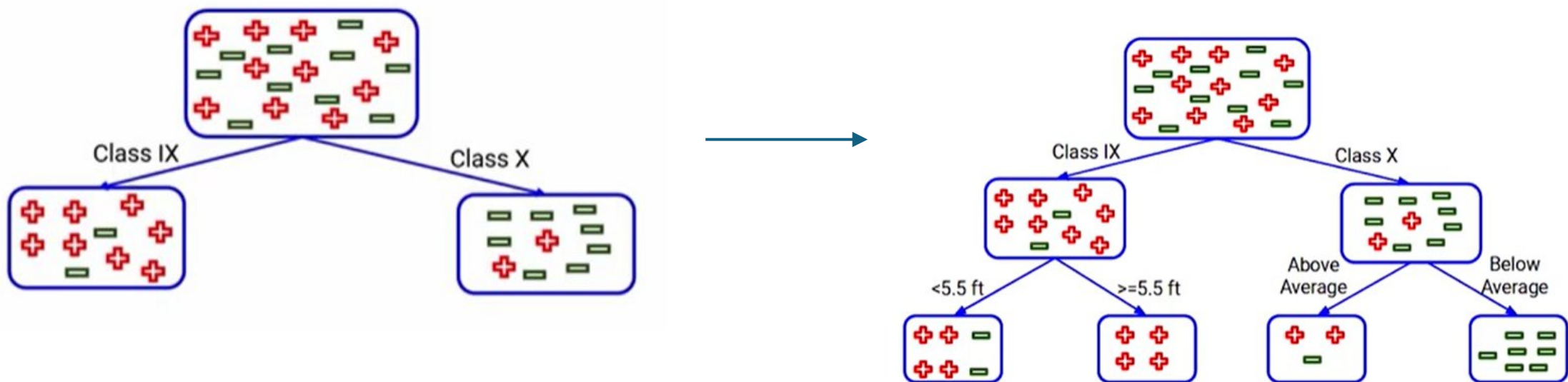
Which is better split, DT should be able to separate the classes , last on Split is pure nodes

- purity in DT



Very rare that completely different class can be obtained from pure nodes after the split, Ideal case is where features gives completely pure nodes,

Objective of DT is to have pure nodes



DT don't stop at first split, can have more split

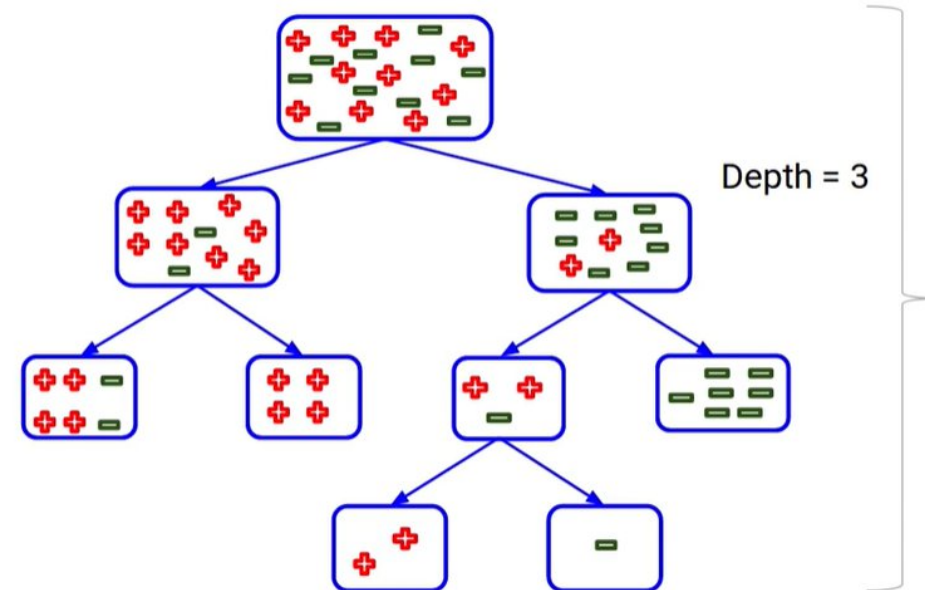
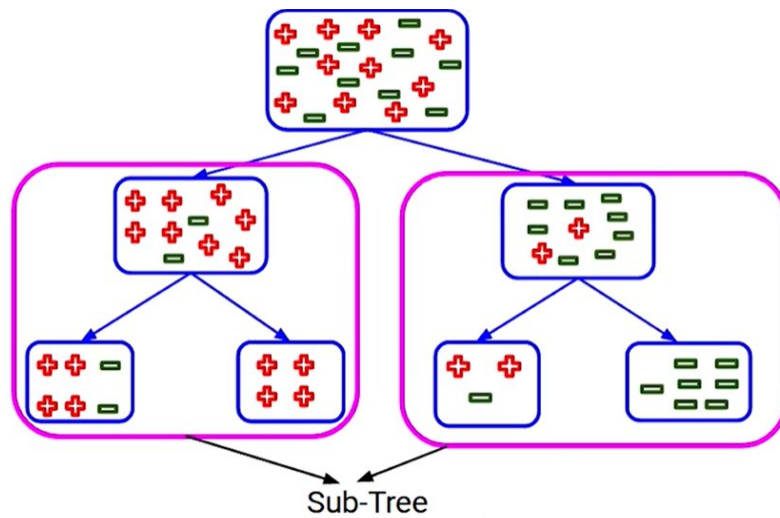
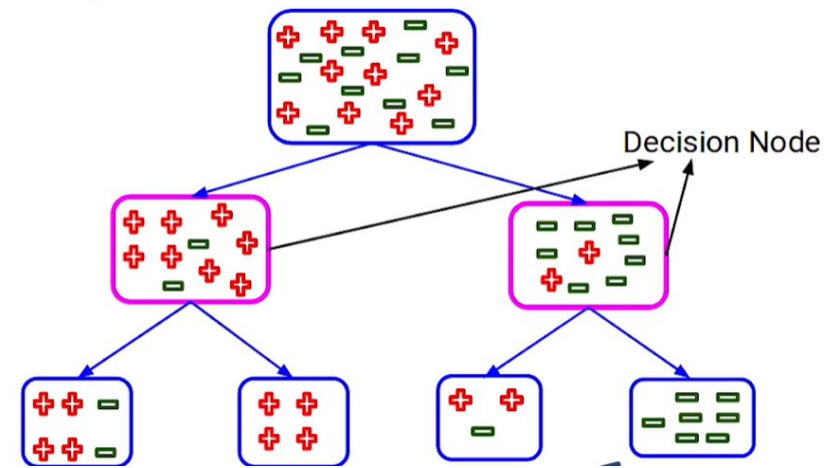
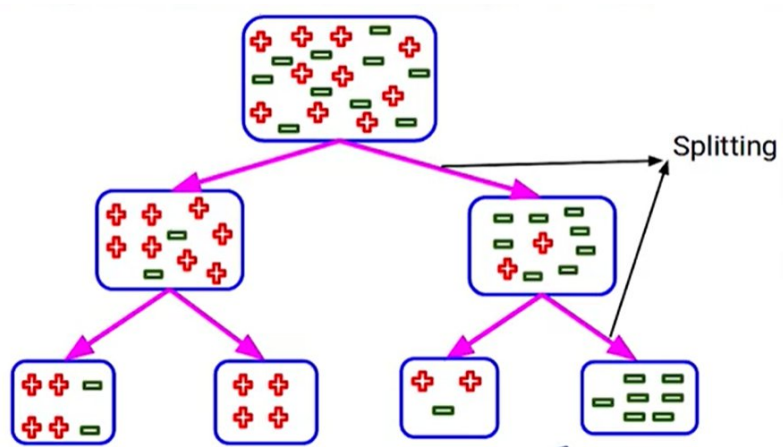
We can have multiple split, multiple decisions in DT, so how to decide which split is better or what should be the first split and second split etc

What should be sequence split made.

Techniques are there to decide purity of nodes based on which feature can decide best splitting point, also how to select 5.5 or 6 in height feature

How to select best split point in DT

1. Decision tree split the nodes on all available variables
2. select the split which results in most homogenous sub nodes



Techniques used to select which is best split

- **Gini impurity** – calculate the impurity of nodes , values ranges form 0-1 , if gini value is higher then low will be the purity of nodes, works on categorical targets, only performs binary splits

Calculate the Gini impurity for sub-nodes:

- *Gini Impurity = 1 - Gini*
- Gini = Sum of squares of probabilities for each class/category

- $$\text{Gini} = p_1^2 + p_2^2 + p_3^2 + \dots + p_n^2$$

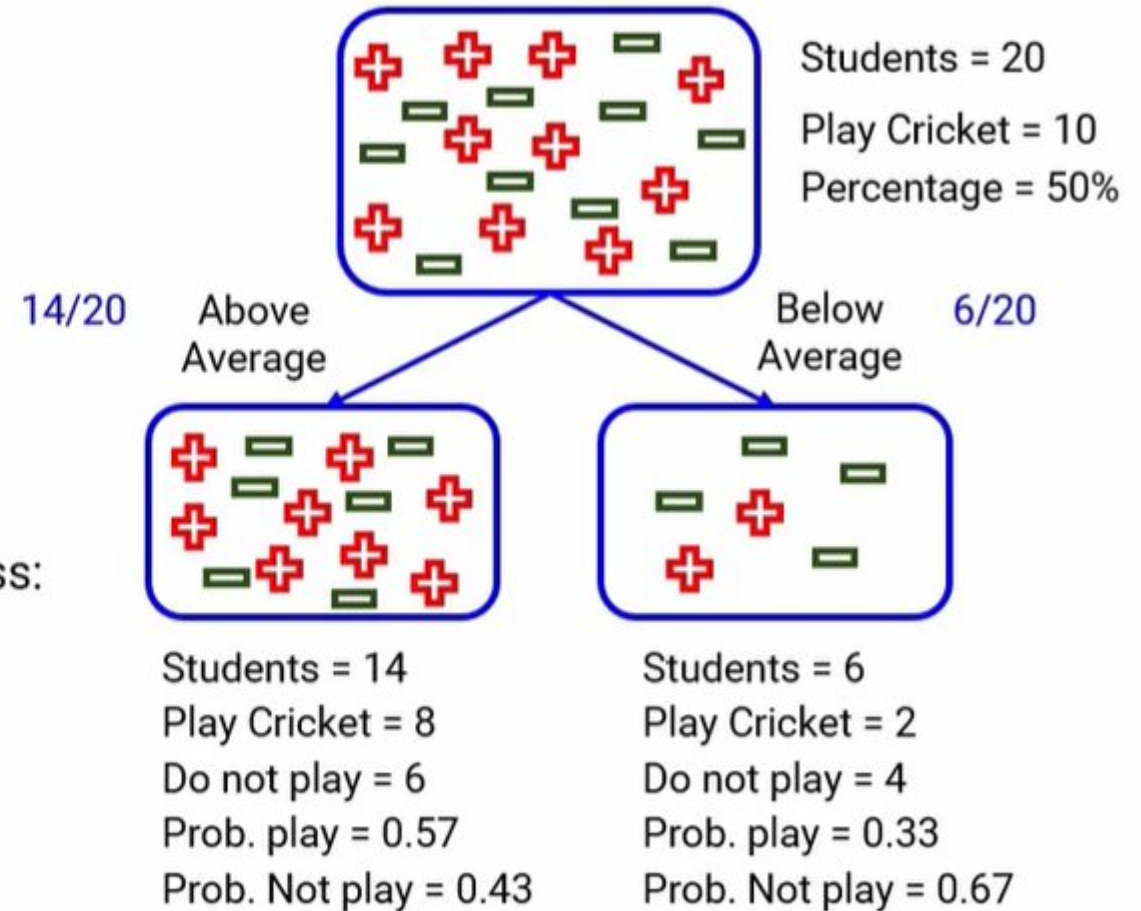
- Therefore,

- $$\text{Gini Impurity} = 1 - \left(p_1^2 + p_2^2 + p_3^2 + \dots + p_n^2 \right)$$

- **CHAID Decision** Tree Chi-Square Automatic Interaction Detection
- **Reduction in variables**

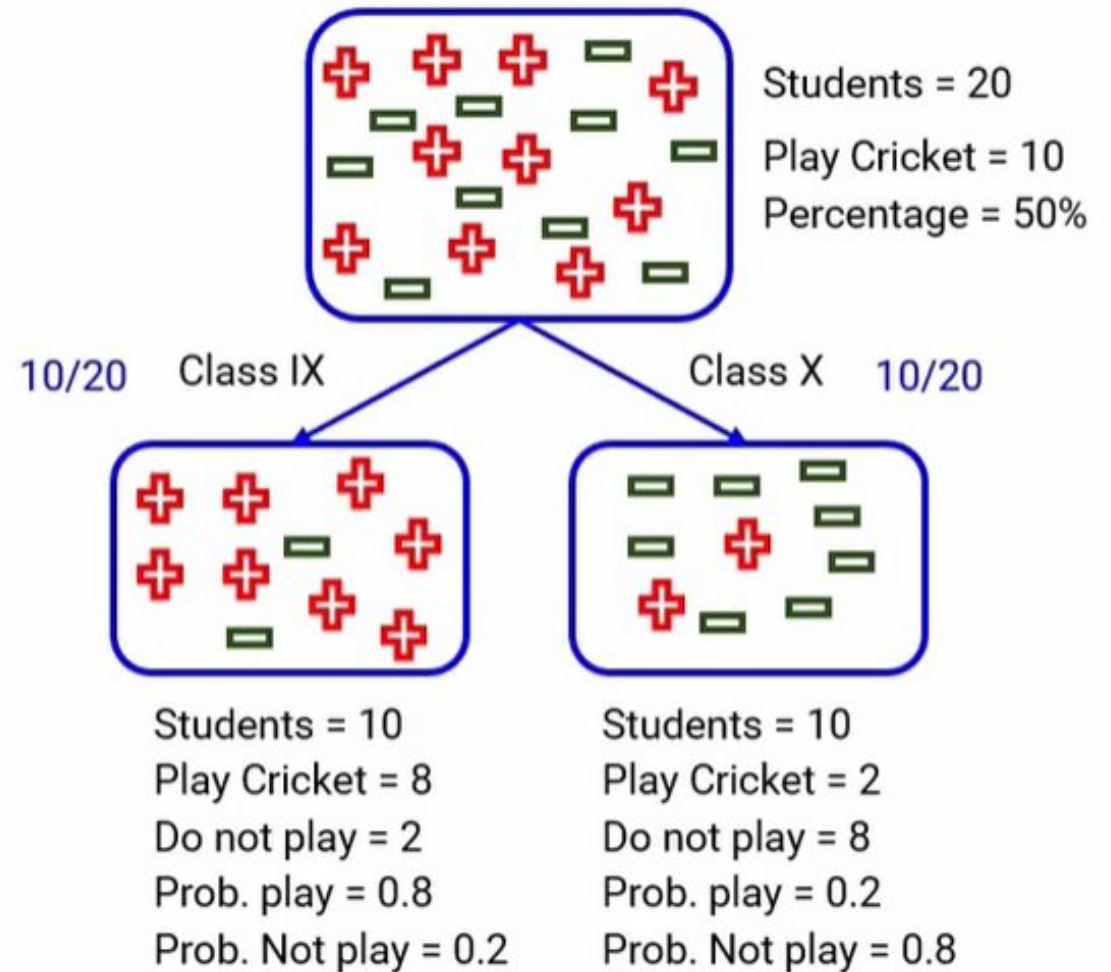
Split on Performance in Class

- Gini Impurity: sub-node Above Average:
 $1 - [(0.57)*(0.57) + (0.43)*(0.43)] = 0.49$
- Gini Impurity: sub-node Below Average:
 $1 - [(0.33)*(0.33) + (0.67)*(0.67)] = 0.44$
- Weighted Gini Impurity: Performance in Class:
 $(14/20)*0.49 + (6/20)*0.44 = 0.475$



Split on Class

- Gini Impurity: sub-node Class IX:
 $1 - [(0.8)*(0.8) + (0.2)*(0.2)] = 0.32$
- Gini Impurity: sub-node Class X:
 $1 - [(0.2)*(0.2) + (0.8)*(0.8)] = 0.32$
- Weighted Gini Impurity: Class:
 $(10/20)*0.32 + (10/20)*0.32 = 0.32$



Split

Performance in Class

Class

Weighted Gini Impurity

0.475

0.32

- CHAID (Chi-Square Automatic Interaction Detection) is a decision tree algorithm used mainly for classification. It builds the tree by selecting the feature that has the strongest statistical association with the target variable using the Chi-square test.

Play Tennis dataset

Day	Outlook	Temp.	Humidity	Wind	Decision
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

We need to find the most dominant feature in the data set.

Outlook feature

Outlook feature has 3 classes: Sunny, Rain, and Overcast. There are 2 decisions: Yes and No.

We first find the number of Yes decisions and No decisions for each class.

Outlook	Yes	No	Total	Expected	Chi-square (Yes)	Chi-square (No)
Sunny	2	3	5	2.5	0.316	0.316
Overcast	4	0	4	2.0	1.414	1.414
Rain	3	2	5	2.5	0.316	0.316

Expected = total/2

$$\sqrt{\frac{(2 - 2.5)^2}{2.5}} = 0.316$$

Chi-

Square(Outlook)=0.316+1.414+0.316+0.316+1.414+0.316

$$\text{Chi-Square}(S, Y) = \sqrt{\frac{(E - O)^2}{E}}$$

S= sunny given Y=yes

Where:

• χ^2 = Chi-Square statistic, O = Observed frequency, E = Expected frequency

• Σ = Sum of the values for all categories

Chi-square value of Outlook is the sum of Chi-square Yes and Chi-square No columns.

$$0.316 + 0.316 + 1.414 + 1.414 + 0.316 + 0.316 = 4.092$$

Now Chi-square values for other features.

Temperature Feature

This feature has 3 classes: Hot, Mild, and Cool. The following table summarizes the Chi-square values for these classes.

Temperature	Yes	No	Total	Expected	Chi-square (Yes)	Chi-square (No)
Hot	2	2	4	2	0	0
Mild	4	2	6	3	0.577	0.577
Cool	3	1	4	2	0.707	0.707

Chi-square Value of Temperature Feature

$$0 + 0 + 0.577 + 0.577 + 0.707 + 0.707 = 2.569$$

Humidity Feature

- Humidity has 2 classes: High and Normal. The following table summarizes the Chi-square values.

Humidity	Yes	No	Total	Expected	Chi-square (Yes)	Chi-square (No)
High	3	4	7	3.5	0.267	0.267
Normal	6	1	7	3.5	1.336	1.336

Chi-square Value of Humidity Feature

$$0.267 + 0.267 + 1.336 + 1.336 = 3.207$$

Feature	Chi-square Value	
Outlook	4.092	Since Outlook has the highest Chi-square value (4.092), the CHAID Decision Tree selects Outlook as the root node (most dominant feature) for the first split.
Temperature	2.569	
Humidity	3.207	
Wind	1.604	

Steps for Developing a CHAID Tree

1. Start with complete training data in the root node.
2. Check the statistical significance of each independent variable.
3. The variable with the least p-value, based on the statistical tests described in Step 2, is used for splitting the data set, thereby creating subsets.
4. Using the independent variables, repeat Step 3 for each of the subsets of the data until:
 1. a) All the dependent variables are exhausted or they are not statistically significant at α (alpha).
 2. b) The stopping criteria are met.
5. Generate business rules for the terminal (leaf) nodes of the tree.

Algorithm which decide best fit for when dealing with continuous target variables like taxi fare, sales etc

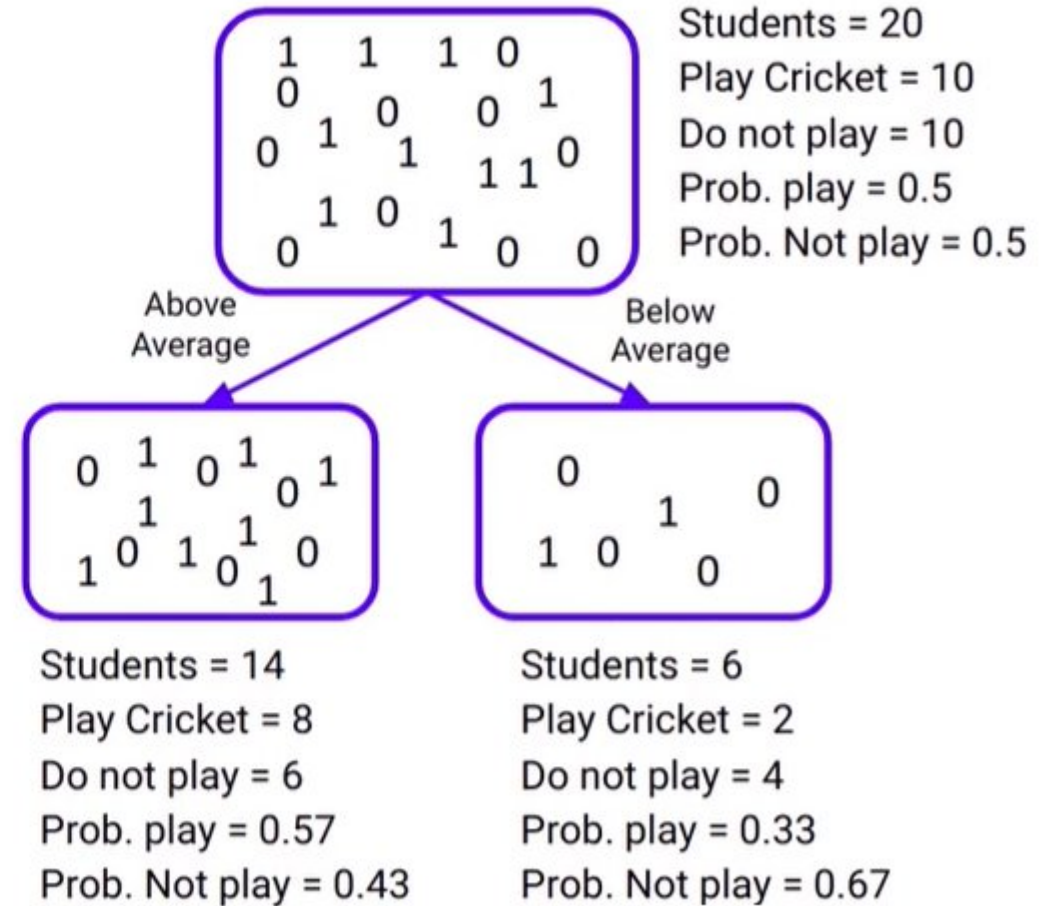
- reduction in variables – split with lower variance is selected
- Steps : Calculate the variance of each child node, calculate the variance of each split as the weighted average variance of each child node.

- $$\text{Variance} = \frac{\sum (X - \mu)^2}{n}$$
 where:

- X = Individual data value
- μ = Mean of the data
- n = Total number of observations
- $\sum$ = Summation of squared differences from the mean

Plays Cricket = 1 Do not play Cricket = 0

- Above Average node:
 - Mean = $(8*1 + 6*0) / 14 = 0.57$
 - Variance = $[8*(1-0.57)^2 + 6*(0-0.57)^2] / 14 = 0.245$
- Below Average node:
 - Mean = $(2*1 + 4*0) / 6 = 0.33$
 - Variance = $[2*(1-0.33)^2 + 4*(0-0.33)^2] / 6 = 0.222$
- Variance: Performance in Class:
 $(14/20)*0.245 + (6/20)*0.222 = 0.238$

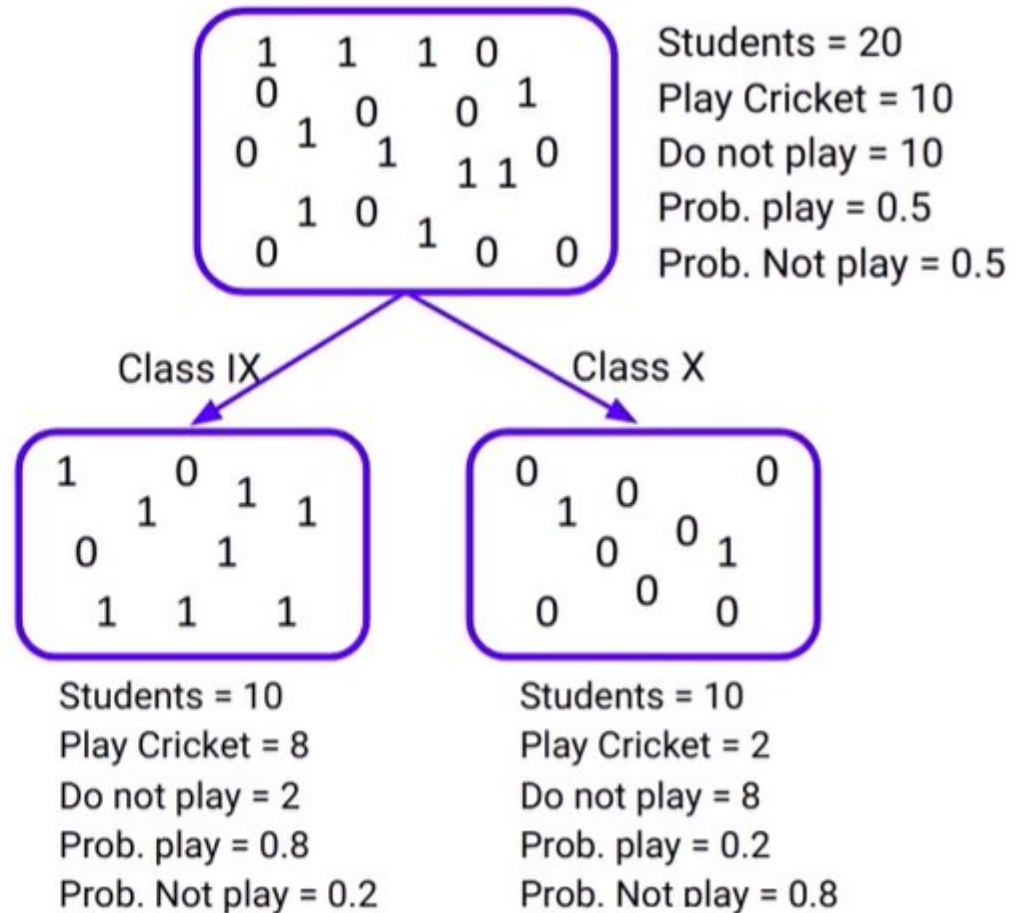


- Class IX node:
 - Mean = $(8*1 + 2*0) / 10 = 0.8$
 - Variance = $[8*(1-0.8)^2 + 2*(0-0.8)^2] / 10 = 0.16$

- Class X node:
 - Mean = $(2*1 + 8*0) / 10 = 0.2$
 - Variance = $[2*(1-0.2)^2 + 8*(0-0.2)^2] / 10 = 0.16$

- Variance: Class:

$$(10/20)*0.16 + (10/20)*0.16 = 0.16$$



Split

Performance in Class

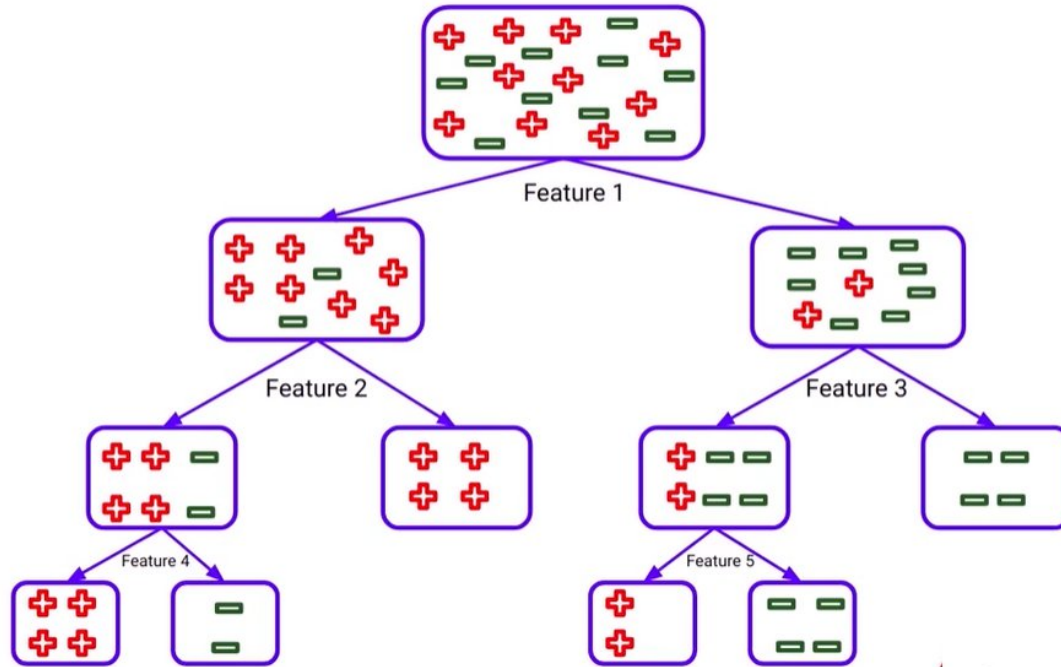
Class

Variance

0.238

0.16

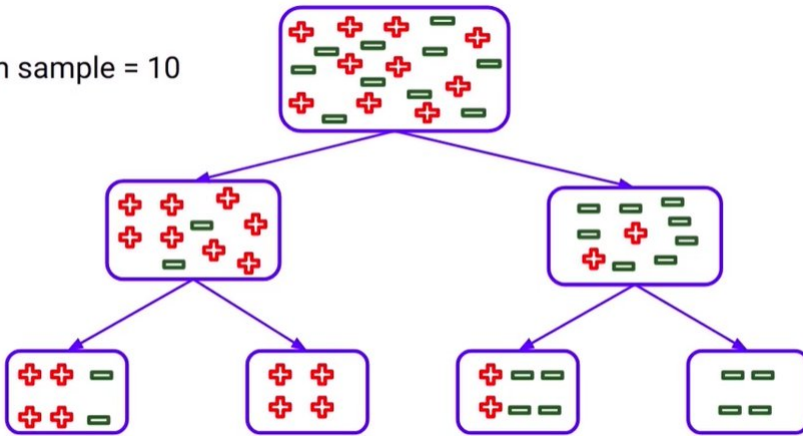
Optimizing performance of DT



If no constraint is kept on tree it grows until it complete all pure nodes , model fits on the training observation and get 100 % accuracy in training .
So if we train unseen data model leads to overfitting

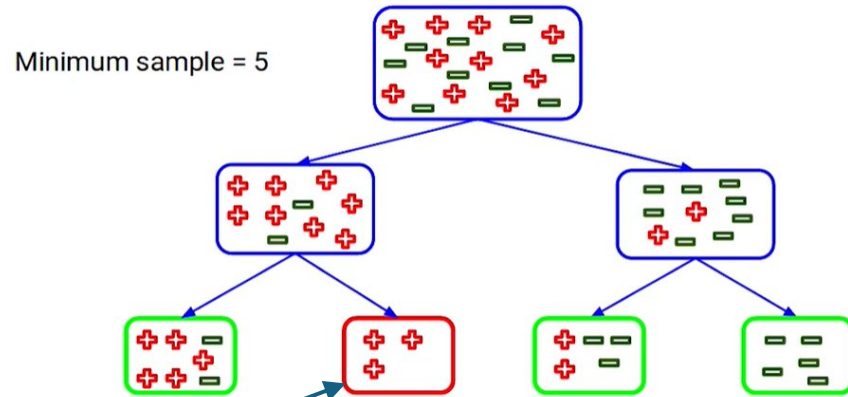
- sol: set a threshold on growth of tree. Depth=1 in such cases training data also won't be good enough for accuracy checking and leads to underfitting
 - So both overfitting and underfitting need to be solved for DT
1. Set minimum samples for a node split

Minimum sample = 10



- a. Higher values control overfitting.
- b. Too high values can lead to underfitting. Model don't learn anything

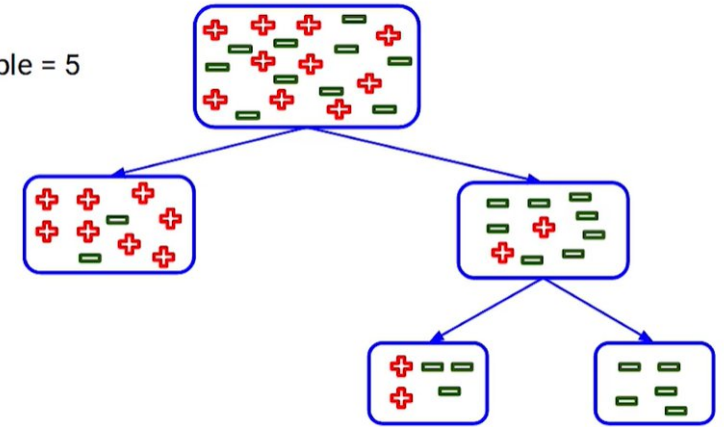
2. set minimum sample for terminal nodes



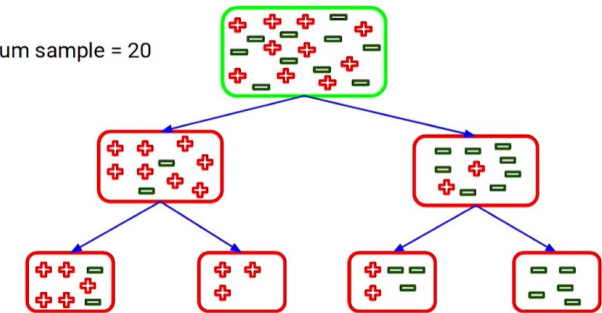
So parent node become
leaf node

- a. Higher value controls overfitting.
- b. Too high values can lead to underfitting.

Minimum sample = 5

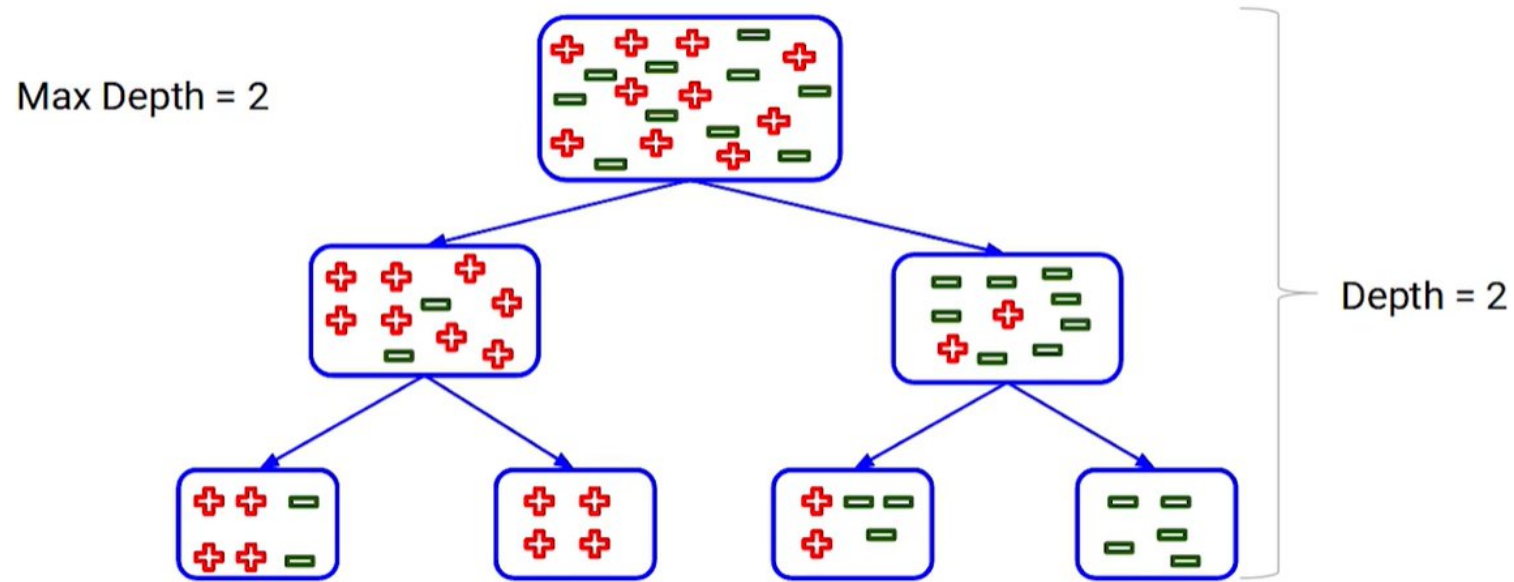


Minimum sample = 20



Here we cant split further form root node
itself cause model to not learn anything
so choose value which is not too low
and not too high

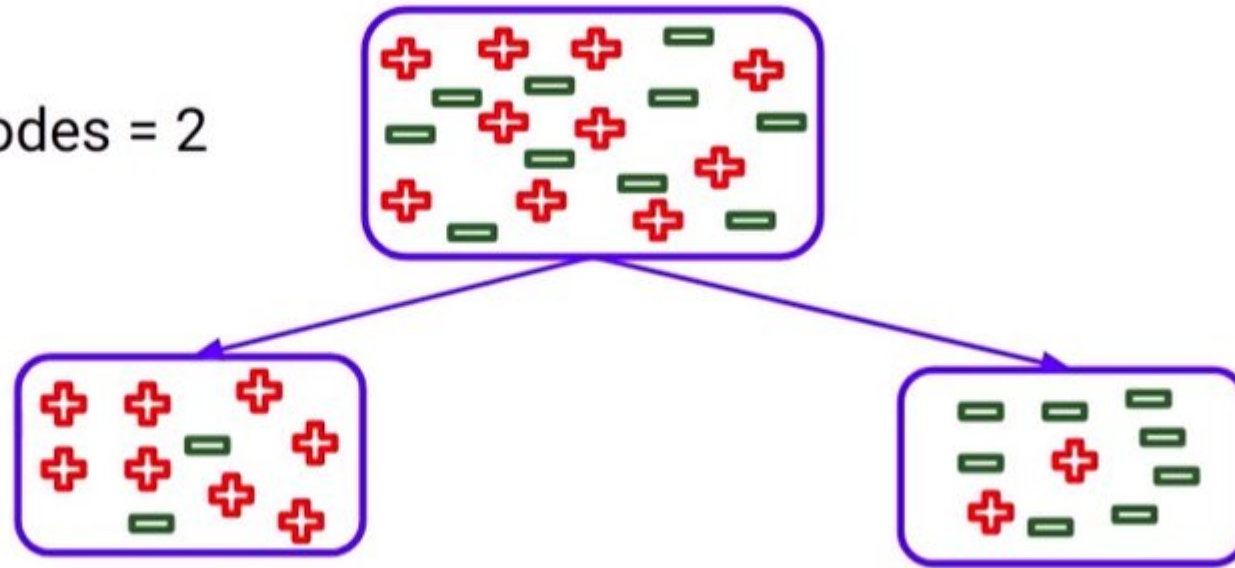
3. Set maximum depth of tree



- a. Higher depth can lead to overfitting.
 - b. Lower depth can lead to underfitting.
- So depending on problem set the max depth

4. Maximum number of terminal nodes

Max Terminal nodes = 2



Consider the dataset below.

ID	Age	Income	Student	Buy Laptop
1	Young	High	No	No
2	Young	Low	Yes	Yes
3	Middle	High	Yes	Yes
4	Senior	Low	Yes	Yes
5	Senior	High	No	No
6	Middle	Low	No	Yes

Answer the following:

1. Calculate the Gini Impurity of the root node.
2. Compute the weighted Gini Impurity for all three features.
3. Which feature will GINI select as the first split? Explain your reasoning.
4. Which feature is likely to be selected by CHAID? Explain your reasoning using the Chi-square criterion.